

Advanced Dual-Modality Predictive Modelling for Hepatic Pathology

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Abstract— The main aim of this research paper is to predict liver diseases with high accuracy. The prediction is done by applying various machine learning and deep learning techniques which can be applied to both numeric and image datasets. Liver diseases are not easy to diagnose at early stages. This has become a significant health concern worldwide. The liver disease dataset is taken from the Kaggle repository. The preprocessing of dataset is done to handle missing values, outliers, and convert categorical attributes into numerical ones. The numeric dataset is then analyzed using multiple machine learning algorithms, including Random Forest, Adaboost, and Gradient Boosting, to predict the presence of liver disease. Medical images are processed using deep learning architectures like Visual Geometry Group Network and ResNet (Residual Network). These architectures are trained to extract relevant features from liver images to make disease predictions. To determine the total prediction accuracy, the models' output is pooled after they have been trained independently on the two datasets. Models undergo extensive testing to guarantee that they produce accurate and dependable predictions, and their performance is assessed using measures such as accuracy, precision, recall, and F1 score.

Keywords—Random Forest, Adaboost, Gradient Boost, Visual Geometry Group Network, ResNet

I. INTRODUCTION

Numerical and imaging information are combined through several phases in the holistic approach to liver disease prediction. The first dataset utilized comes from the UCI repository and is called the Indian Liver Patient Dataset. Most of the ten qualities are numerical. During preprocessing procedures, the non-numerical properties are transformed into numerical values. During the preprocessing stage, the data is converted from categorical attributes into numerical values. To guarantee that the data is in a format that is appropriate for machine learning algorithms, normalization and scaling procedures are applied to numerical attributes. The model's performance can then be accurately assessed by dividing the dataset into training and testing subsets. For the numeric data, three machine learning algorithms are employed: Random Forest, Adaboost, and Gradient Boosting. These algorithms

are trained on the numeric attributes to predict liver disease. Simultaneously, image data is processed using two well-known deep learning architectures: VGG (Visual Geometry Group Network) and ResNet (Residual Network). These models are trained on the image datasets to extract relevant features and make predictions. Following the separate training on numeric and image data, the models' results are combined to evaluate their collective accuracy. This stage assesses how well the integration of numeric and image data improves prediction accuracy.

Patients and illnesses lacking specific characteristics are frequently left out of the data set of the current system. Most of these systems were created for more severe conditions like heart disease, cancer, etc. Results may occasionally be inaccurate because the pre-selected characteristics did not account for changes in the disease and its influencing factors. Because of the environment we live in, the symptoms of illnesses also fluctuate with time. Additionally, the majority of systems in use today need users to complete drawn-out surveys, which prolongs user wait times. This includes both picture and number datasets in our system. In the first, three methods are applied to a numerical dataset, nine qualities are chosen, and the approach with the highest accuracy is chosen. The liver disease will be predicted by giving the characteristics values. Two algorithms and a liver picture are included in the second dataset. After then, the dataset is divided into subsets for testing and training, enabling a precise assessment of the model's performance. Three machine learning algorithms—Random Forest, Adaboost, and Gradient Boosting—are used for the numerical data. To forecast liver illness, these algorithms are trained using numerical characteristics.

II. LITERATURE SURVEY

There are many techniques to predict the liver disease which has lot of benefits and negativity. The fuzzy logic techniques are used to handle the imperfect data to provide the valid insights. The algorithms like data mining, SVM and Naive

bayes produces different results with variable accuracy. The decision tree algorithms can be used to produce the results with least variance.

It is very necessary to adopt proper strategies as highlighted in the survey. This has a major effect on the performance. The more reliable diagnostic tools can be used to produce the more accurate results. As highlighted in the survey, each work has added significant contribution in the prediction of the liver disease. The work has been carried out using fuzzy logic for the data collected from patients with liver cirrhosis using child-pugh classification[1].

The fuzzy logic strategy is chosen for better performance in the proposed technology. This method is used as it deals with the uncertainty and also enhances the child-pugh classification. The fuzzy method provides more adaptable environment as increasing liver cirrhosis severity. This method models the complex problems using binary logic.

The complexity and interpretability issues are addressed. The algorithm performance is evaluated by accuracy measures and comparing integrated model results. Additionally, by integrating visual and numerical data, it offers a more thorough diagnostic tool that gets around the limitations of domain-specific expertise needed for fuzzy logic systems.

In the work, the application of various decision tree algorithms for the early diagnosis of liver disease is examined [2]. The decision tree algorithms that are assessed in this study are J48, LMT, Random Forest, Random Tree, REP Tree, Decision Stump, and Hoeffding Tree.

Decision tree algorithms were proposed for the diagnosis and prediction of excessive alcohol-induced liver disease using data mining techniques [3]. Training and testing the dataset evaluates the model's performance. To tackle issues like as data completeness and quality, utilize data pretreatment methods and utilize a large dataset from Kaggle and To increase accuracy and robustness, combine numerical and visual data. Utilize deep learning on picture data to find visual indicators of liver illness, improving early identification of alcohol-related liver damage and providing a noninvasive diagnostic alternative.

The effectiveness of the Support Vector Machine (SVM) and Naive Bayes Classifier (NBC) algorithms for predicting liver illness is compared[4]. It has High precision for intricate datasets, but computationally costly, sensitive to noise, which could cause incorrect classifications. Although it is quicker and easier, it makes the unreasonable assumption that features are independent for medical data. inadequate capacity to manage intricate feature interactions. Use alternative machine learning methods like Random Forest, Adaboost, and Gradient Boost to address this challenge. These algorithms are better at generalizing to numerical datasets, minimize noise sensitivity, and provide advantages over SVM and NBC.

Various machine learning algorithms and data mining techniques employ the Adaptive Neuro-Fuzzy Inference System (ANFIS), Decision Tree (DT), and K-Nearest Neighbor (K-NN) to diagnose liver diseases [5]. These

methods are evaluated for specificity, accuracy, sensitivity, and precision using MATLAB. ANFIS is the most accurate method, however it depends a lot on the quality of the data. DT and K-NN algorithms are less accurate despite being simpler. To overcome this challenge, integrate numerical and visual data to reduce dependence on a single source, and use trustworthy data processing and cleaning techniques. establish an all-encompassing framework to increase forecast accuracy and early intervention.

According to Prediction of Different Types of Liver Diseases Using Rule-Based Classification Model [7], the requirement for many laboratory tests usually complicates the diagnosis procedure. The lack of investigation into liver function test anomalies by general practitioners may postpone the identification of liver diseases, which can have major health consequences if left untreated.

decision trees are vulnerable to instability. We intend to use effective machine learning methods to improve accuracy in order to combat overfitting and instability. For instance: • 75% to 90% gradient boosting • 70% to 90% for Random Forest AdaBoost: 70% to 85%. • CNN: 85% to 95% (ResNet, VGG). The study makes use of the Adaptive Neuro-Fuzzy Inference System (ANFIS), Decision Tree (DT), and K-Nearest MATLAB is used to assess the accuracy, sensitivity, precision, and specificity of these techniques. Multiple laboratory tests are frequently needed to diagnose liver illnesses, which makes the process more difficult. General practitioners could not look into liver function test abnormalities enough, which could delay the detection of liver illnesses, which, if unchecked, can have serious health repercussions. A dataset with 12 characteristics from 583 patients—441 men and 142 women—was created. The comparison highlights how important it is to select the appropriate strategies based on the specific problem context and performance requirements. By providing valuable insights into the challenging task of liver disease prediction, each study contributes to the development of more accurate and reliable diagnostic tools.

III. SYSTEM DESIGN AND ANALYSIS

A high-level diagram provides an overview of the main components and interactions within the software development project. This visual representation helps stakeholders understand the project's structure and flow from initial requirements gathering to final implementation.

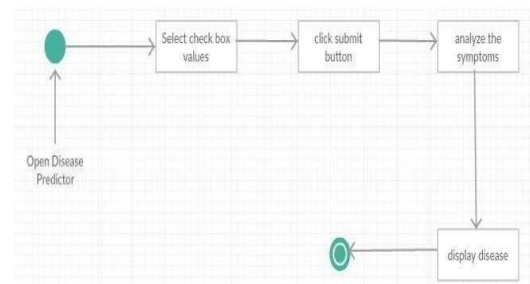


Fig1.High Level Design

Regardless of how they will be met, the system's requirements are established during the analysis stage. This stage defines the problem that the customer is trying to solve. At the end of this phase, the final product is a requirement document. Making an accurate, thorough, consistent, and verifiable model of the system is the aim of analysis. The Analysis model is the name given to this concept. A conceptual model that describes a system's structure, behavior, and other features is called a system architecture. An architecture description is a formal description and representation of a system that is organized to make it easier to reason about its structures and actions.

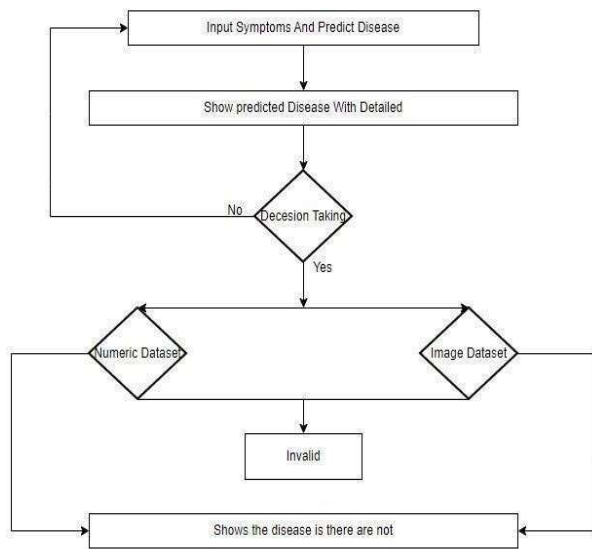


Fig2. System Architecture

Algorithm

- Stage 1: Gather the numeric and image dataset and group them into a single report for easy transit.
- Stage 2: Pre-process the data.
- Stage 3: Split the data for both train and test datasets.
- Stage 4: In numeric there is using of three algorithms, they are Random Forest, Adaboost, Gradient Boost.
- Stage 5: In image there are of two datasets and they are VGG, ResNet
- Stage 6: combining of these both we will find the accuracy of the algorithms.
- Stage 7: In this both numeric and image datasets will be trained and tested.
- Step 8: Make the note of accuracy.
- Step 9: Check the accuracy of the model.
- Step 10: Stop here if the precision is good and calculate the exactness rate.

The goal of the coding or programming phase is to convert the system's design, which is created during the design phase, into code in a specific programming language that a computer can run and that carries out the computation that the design specifies. Both testing and maintenance are impacted by the coding stage. Coding should be done to lower the cost of subsequent stages, not to lower the cost of

implementation. Programming style refers to the set of guidelines that a programmer must adhere to in order to satisfy coding attributes like extensibility, traceability, understandability, and modifiability. We named the variables and procedures in the existing system according to the coding guidelines.

Verification is the process of ensuring that the product is constructed correctly. The process of determining if the correct product is constructed is known as validation. Throughout the system's development, the object's coding was carefully examined from a variety of angles, including how it was incorporated and how it was designed. Each part of the suggested system has been validated by providing distinct input data while each module is operating.

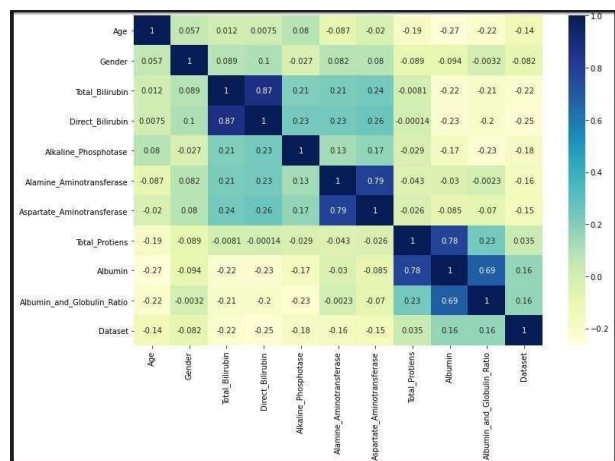


Fig3.Attributes (Numeric)

IV. RESULTS

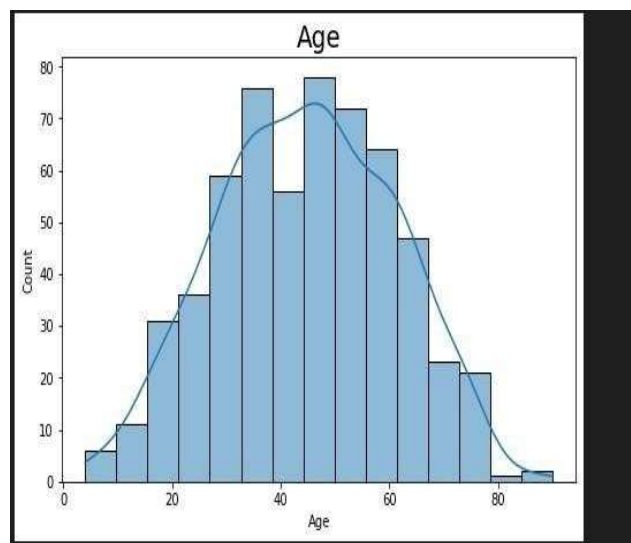


Fig4. Image Liver Disease Output



Fig5. Image Liver Disease Output

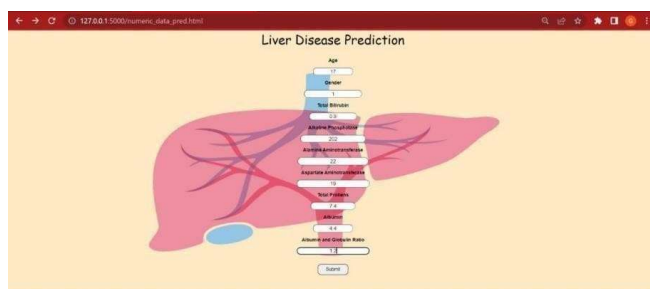


Fig6. Image Liver Disease Output

Sl.No	Test Case	Description	Expected Result	Actual Result	Pass/Fail
1	Load the numeric values	Giving numeric values to text boxes on the website	Liver Disease is present from numeric values is predicted	Liver disease is present is predicted	PASS
2	Load the numeric values	Giving numeric values to text boxes on the website	Liver Disease not present from numeric values is predicted	Liver disease not present is predicted	PASS
3	Image consists of Liver Disease	The image consists of values will be given to algorithm to predict the disease	Liver Disease is present	Liver Disease is detected in image	PASS

Table1. TestCases

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